

Digital Audio Player — Project Brief

Rev 3 | Scoping and architecture | Alexander Loker

A from-scratch portable digital audio player: schematic, PCB, simulation, firmware and enclosure all designed personally. The device is a genuine end product — small, pleasant to use, and measurably good — but the primary return is closing two deliberate skill gaps: **custom MCU board design** and **analog / mixed-signal design backed by simulation**.

Changes from Rev 2: charger selected (TI BQ24250RGET). The ADI-only power constraint has been narrowed — LTspice-native silicon is now required only where simulation drives the design (boost, inverting charge pump, analog LDOs), and the charger is exempt because its behaviour is a state machine that SPICE does not usefully characterise. Charger design values, TS network requirements and thermal figures added.

1. Objectives

#	Objective	How it is satisfied
1	Own an audio player worth using	Portable microSD player, FLAC and MP3, single-ended 3.5 mm output tuned for IEMs, 2.4 inch display.
2	Close the MCU-board knowledge gap	Custom ESP32-S3 board from the module up: power tree, boot straps, SDMMC, USB, I2S, I2C charger control.
3	Work with data converters	ES9038Q2M current-output DAC with a discrete I/V and filtering chain designed from scratch.
4	Work with analog signals	I/V conversion, differential summing, reconstruction filtering, gain structure, output impedance control, low-noise power conditioning.
5	Simulate where simulation earns its place	LTspice for the boost, charge pump and LDOs; TI PSpice / TINA-TI for the signal chain. See Section 6.
6	Learn	Phased revisions isolating one class of risk at a time, each ending in a measurement.

2. Locked Scope

Decision	Choice	Rationale
MCU	ESP32-S3	USB-OTG device peripheral, 4-bit SDMMC, dual-core 240 MHz, octal PSRAM. ESP-IDF / FreeRTOS.
DAC	ES9038Q2M	Differential current output; internal DPLL rejects incoming I2S jitter. Current output forces the full analog chain.
Source	microSD, local files only	No Bluetooth, no streaming, no USB Audio Class.
Formats	FLAC 16/44.1 minimum, MP3	Decoded on the MCU to PCM. The DAC never sees a file format.
USB-C role	Charging + file transfer only	TinyUSB MSC exposes the microSD as a drive.
Output	3.5 mm TRS, single-ended, DC coupled	Balanced dropped. DC coupling preserves the sub-1-ohm target but requires offset protection.
Load target	16-32 ohm IEMs	Unforgiving of noise floor and output impedance.
Output amp	TI OPA1622, 10-pin VSON (DRC)	Dual channel. Thermal pad connects to V–, not ground. EN pin used for shutdown.

Decision	Choice	Rationale
Analog rails	$\pm 3.3\text{ V}$	Supply range is $\pm 2\text{ V}$ to $\pm 18\text{ V}$ on (V+)-(V-). Gives $\sim 1.9\text{ V}$ headroom at peak output.
Display	2.4 inch IPS, 240x320, SPI	Glass ($\sim 43 \times 60\text{ mm}$) fits inside the battery footprint. ST7789 or ILI9341. No touchscreen.
Battery	Jauch LP605060JU+PCM, 1S1P	3.7 V, 1850 mAh, $63.0 \times 51.0 \times 6.6\text{ mm}$ actual envelope, integrated PCM, 70 mm flying leads, no internal NTC .
Charger	TI BQ24250RGET	3 MHz synchronous buck with PowerPath, VQFN-24 $4 \times 4\text{ mm}$. I2C (0x6A) plus standalone fallback. TS pin gives hardware JEITA. Charge at 500 mA (0.27C). Active production status verified.
Charger variant choice	'250 spin specifically	The bq24250 carries EN1/EN2 rather than D+/D-, so it does not contend for the USB data lines. The bq24251 is NRND; the bq24253 is standalone-only.
Fuel gauge	Voltage-based, no coulomb counter	Load-compensated lookup table characterised from the spare cell. Requires ESP32-S3 ADC calibration and a gated divider.
Power silicon rule	LTspice-native where simulation drives design	Applies to the boost, inverting charge pump and analog LDOs. Charger exempt — its CC/CV state machine, termination and thermal foldback are not SPICE-characterisable. TPS63020 reuse remains excluded because the boost <i>is</i> simulation-driven.
Signal-chain sim	TI PSpice for TI / TINA-TI	TI models will not import into LTspice.
EDA / CAD	Altium (AL-PN library), SolidWorks	Existing toolchain carries over.

3. Explicitly Out of Scope

Item	Status	Note
4.4 mm balanced output	Dropped	Would require four amp channels.
Bluetooth / A2DP	Dropped	ESP32-S3 is BLE-only.
USB Audio Class	Deferred	Full-speed USB caps UAC1 around 24-bit / 96 kHz. Revisit at Rev B.
Battery telemetry / coulomb counting	Rejected	Voltage-based gauging adopted. The charger provides fault and loop status over I2C but no battery-voltage ADC.
BC1.2 / USB-C PD negotiation	Rejected	Input current limit set by EN1/EN2 pins or I2C. 5.1 kohm CC pull-downs only.
Touchscreen	Rejected	Encoder and buttons are better in-pocket.
Line-in / recording	Deferred	Nothing in the design forecloses it.
DSP / parametric EQ	Deferred	Add in firmware once the signal path is trusted.
Bare-metal firmware	Rejected	Not where the learning value is.
Vented enclosure / active cooling	Rejected	Heat reaches the cell by conduction, not convection; venting would not interrupt that path and would admit dust and moisture. A fan is also an acoustic noise source in a listening device. Made moot by the switching charger — see Section 4.6.

4. System Architecture

Signal chain: microSD → SDMMC (4-bit) → ESP32-S3 (decode to PCM in a PSRAM ring buffer) → I2S → ES9038Q2M → I/V conversion → differential summing and MFB low-pass → OPA1622 output stage → 3.5 mm TRS. A local audio oscillator feeds the DAC MCLK directly.

Power chain: Li-Po cell → BQ24250 (charge + PowerPath, SYS output) → boost to ~5 V → digital buck branch at 3.3 V, and an analog branch through LT3042 / LT3093 to ±3.3 V. One uninterrupted ground plane; separation by pour partitioning, never a split plane.

4.1 Digital subsystem

- ESP32-S3 with octal PSRAM. Decode on one core, UI and I2S service on the other.
- SDMMC in 4-bit mode. Ring buffer sized against worst-case card latency; instrument minimum occupancy. A load switch on card VDD allows power-gating between bursts — tristate host pins before gating, since power must reach VDD before any I/O pin goes high.
- TinyUSB MSC. Enumeration must stop playback and release the FAT volume before host mount. **The USB data lines are exclusively the S3's** — the bq24250 spin has no D+/D− pins.
- FLAC and MP3 decode entirely on the MCU. Format support costs cycles and RAM only.
- UI: 2.4 inch SPI TFT, EC11 encoder, tactile switches. Partial updates only — a full 240x320 frame at 16 bpp is 153.6 kB, about 31 ms at 40 MHz SPI.
- Display off during playback with wake on interaction. A power requirement, not a preference.
- **Charger control:** I2C at address 0x6A, plus two GPIOs to EN1/EN2 for the input current limit and one to /CE. Charger STAT and INT are open-drain into GPIOs through 10 kohm pull-ups.

4.2 Clocking

Jitter-limited SNR is $-20 \log_{10}(2 \pi f t_j)$: roughly 98 dB at 100 ps and 118 dB at 10 ps, referred to a 20 kHz signal. Periodic jitter from a fractional-N PLL is worse than its RMS figure suggests because it produces discrete sidebands rather than a raised noise floor.

- Two oscillators: 22.5792 MHz (44.1 kHz family) and 24.576 MHz (48 kHz family). Ratio 147/160, so neither derives from the other without a fractional PLL.
- **The ES9038Q2M DPLL means the ESP32-S3 remains I2S master.** A clean MCLK from a local oscillator lets the DAC's DPLL reject jitter on incoming BCK and LRCK. DPLL bandwidth is a tunable, measurable parameter.
- Gate the unused oscillator by its OE pin rather than muxing through logic. **Verify the datasheet states OE tri-states the output** before tying the two together.
- Give the oscillator its own low-noise supply — supply noise becomes phase noise via the supply pushing figure.
- Place the oscillator adjacent to the DAC MCLK pin, 22-33 ohm series termination. Specify by phase noise, not frequency accuracy.

4.3 Analog chain

- The ES9038Q2M presents differential current outputs: an inverting I/V stage per phase (four channels), then differential-to-single-ended summing per channel, then the output driver.
- The I/V stage is a first-order low-pass via its feedback capacitor. The summing stage is a second-order **multiple-feedback (MFB)** section.
- **MFB over Sallen-Key.** Sallen-Key's stopband degrades at high frequency as the op-amp's rising output impedance turns the feedback capacitor into a feedthrough path, limiting rejection to roughly 40-60 dB. MFB keeps rolling off and has a virtual-ground summing node with no common-mode swing.
- **The filter's real job is the delta-sigma modulator's out-of-band shaped noise**, which can be tens of dB above in-band level and extends into the MHz region. Allowing it into the output stage consumes slew rate and intermodulates back into the audio band.
- Target: flat to 20 kHz within 0.1 dB, -3 dB around 60-100 kHz, well attenuated by 1 MHz.
- Opamp allocation: low-noise fast-settling part for I/V (OPA1612 class); OPA1622 for the output stage.
- C0G/NP0 throughout the signal path. Class-II ceramics are piezoelectric and voltage-nonlinear.

4.4 Output stage

$V_{load} = V_{amp} \times Z_{load} / (Z_{out} + Z_{load})$. Multi-driver IEM impedance swings with frequency, so non-negligible Z_{out} produces audible response tilt. **Target under 1 ohm.**

- Closed-loop output impedance is the open-loop value divided by loop gain, so it rises with frequency. Plot it; a single number is meaningless.
- Use **output-inclusive compensation**: audio-band feedback after the isolation resistor, high-frequency feedback before it through a small capacitor.
- **DC coupling requires output offset detection and a muting device** — a DC fault destroys an IEM. The OPA1622 EN pin participates but needs a ramp to avoid pops.
- **Gain structure**: design for ~1 V RMS maximum (1.41 V peak) and attenuate ahead of the output stage. A 110 dB/mW IEM reaches 110 dB SPL from about 179 mV RMS into 32 ohm.
- Copper resistance matters at a 1 ohm budget — 1 oz, 10 mil, 1 inch is about 50 milliohm, and the ground return counts equally. Pour the output nets.
- **OPA1622 thermal pad connects to V-**, so it is a -3.3 V island needing its own pour and via field. Soldering it is required for specified performance.

4.5 Power tree

- **Topology:** cell (3.0-4.2 V) → BQ24250 SYS → boost to ~5 V → LT3042 to +3.3 V; inverting charge pump → LT3093 to -3.3 V. LDO dropout is why the boost is mandatory.
- LT3042 holds high PSRR into the MHz region where switcher energy lives, unlike a conventional LDO. Roughly 0.8 uV RMS output noise over the audio band.
- LTC3265 / LTC3260 to be evaluated as integrated boost / inverter / dual-LDO parts. **Minimum input voltage must be confirmed.**
- Digital and analog take separate branches from SYS so that SD, display and backlight load steps do not modulate the analog rail.
- **Two switchers now sit near the analog section:** the charger at 3 MHz (only while plugged in) and the analog boost (always running). The charger's 3 MHz fundamental is three decades above the audio band and well inside LT3042's effective PSRR range — conducted noise is manageable. Magnetic near-field coupling and ground-return currents are not filterable and must be handled by placement, shielded inductors and return-path control.
- Avoid spread-spectrum modulation on the analog supply path. X5R/X7R DC-bias derating is severe; specify against the derated figure.

4.6 Charging — BQ24250RGET

A 3 MHz synchronous buck charger with dynamic power-path management. Charge current is set to 500 mA (0.27C), which is the bottom of the device's 500 mA to 2 A fast-charge range and gentler than the conventional 0.5C, favouring cycle life.

Parameter	Value	Note
Switching frequency	3 MHz (2.7-3.3)	Three decades above the audio band.
Charge current	500 mA	$R_{SET} = 250 / 0.5 = 500 \text{ ohm}$. Bottom of range.
Input current limit	EN1/EN2 or I2C	500 mA / 100 mA / external RILIM / Hi-Z.
RILIM (if external)	270 ohm for 1 A	$K_{ILIM} = 270 \text{ A}\Omega$.
VBATREG	4.2 V default	Matches the cell. Programmable 3.5-4.44 V in 20 mV steps.
Inductor	1.0 uH, $\geq 2 \text{ A}$ saturation	Shielded, low DCR. Peak ~1.15 A (see below).
Key capacitors	CIN >2.2 uF, CPMID 1 uF, CBOOT 33 nF >15 V, CSYS >22 uF, CBAT >1 uF, CLDO 1 uF	Derate all for DC bias.
I2C address	0x6A	Standard (100 kbps) or fast (400 kbps) mode.
Safety timer	Set to 9 h via I2C	The 6 h default leaves under an hour of margin over a ~4.5-5 h charge. Do not rely on it.
Thermal	~5 degC junction rise	theta-JA 32.9 degC/W (QFN), ~160 mW dissipated at 500 mA. Regulation threshold is 125 degC.

- **PowerPath is the reason for this part.** The charger monitors battery current continuously and reduces charge current when system load exceeds the input limit, so termination works correctly while the device is running. It also starts the system from a deeply discharged or missing battery, and lets the battery supplement system load when the adapter cannot meet peak demand.
- **The switching-versus-linear decision is closed.** A linear charger at 500 mA would dissipate ~0.65 W, roughly 21 degC of rise conducted into a cell sitting 2 mm away, at exactly the moment the cell is climbing toward full charge. The buck dissipates ~160 mW. Heat plus high state of charge is the worst combination for calendar

aging, and no enclosure venting interrupts a conduction path.

- **TS network is mandatory** and replaces the missing cell thermistor. Topology: LDO (4.9 V) → R2 → TS node; NTC and R3 both from TS to ground. Thresholds are **ratiometric** as a percentage of VLDO — 30% hot, 38.3% warm, 56.5% cool, 60% cold — so LDO tolerance cancels.
- R3 flattens the NTC curve when beta is not near 3500 K. Datasheet worked examples: a 10 kohm / beta 4000 K NTC gives R2 = 5 kohm, R3 = 9.82 kohm; a 13 kohm / beta 4500 K gives 5 kohm and 8.8 kohm. **Note the datasheet designates these R2/R3 in Section 9.3.10 but R3/R4 in the Section 10.2.2.1 example — same network.**
- Beta error shifts the VCOOL and VWARM crossings to roughly 13 degC and 47 degC rather than the ideal 10 and 45 degC. Small, but real.
- **On a TS fault, SYS stays active and charge current is disabled** — the device keeps running, charging stops. Exactly the desired behaviour.
- **RISET must always be populated.** Floating ISET with external ISET selected gives zero charge current; shorting it is a fault that suspends charging. Size it for 500 mA so that a watchdog expiry falling back to standalone mode still yields safe defaults.
- **SYSOFF is not a standby mode.** It drops BAT leakage below 1 uA but disconnects the battery from the system entirely, killing the MCU. It is a shipping/storage mode. Normal standby uses the high-impedance leakage path (16 uA typical, 22 uA max).
- Use VDPM: it reduces input current to hold the supply up when a weak source or long cable sags. Short to GND for the 4.68 V default, or divide for a custom threshold. Free protection against poor chargers.
- **PTM mode** turns the part into a plain DC-DC converter with no battery present — useful for bench bring-up before committing a cell to the board.
- USB-C sink compliance still requires 5.1 kohm pull-downs on both CC lines. The part does not negotiate USB-C or PD.
- **Absolute maximum on SYS and BAT is only 5 V** — tighter than intuition suggests. Keep it in mind during bring-up.

4.7 Display integration

- Rev 0 and Rev A use a 2.4 inch breakout module. Rev B integrates a bare panel with an FPC connector.
- Motivation is thickness, not cost. At 2.4 inch the glass fits inside the battery footprint, so the display costs no additional volume.
- New work at Rev B: a 0.5 mm or 0.3 mm pitch FPC connector and tail routing, plus a constant-current boost LED backlight driver (typically 3-4 LEDs in series at 9-20 V forward).
- The backlight driver is a third switcher. Keep it distant from the analog section, use a shielded inductor, and place PWM dimming frequency and harmonics clear of the audio band.
- **Documentation risk:** source from a vendor publishing FPC pinout and backlight configuration.

5. Target Specifications

Parameter	Target	Comment
Output connector	3.5 mm TRS, single-ended	DC coupled, one jack.
Rated load	16-32 ohm	IEMs.
Output impedance	< 1 ohm, 20 Hz - 20 kHz	Most important spec for this load.
Max output	~1 V RMS into 32 ohm	Approx. 30 mW.
Output noise	< 3 uV RMS, A-weighted	Audible hiss floor is the practical limiter.
THD+N	< -105 dB @ 1 kHz, 32 ohm	Rev B goal. The DAC is not the limiting element.
Crosstalk	< -80 dB @ 1 kHz	Single-ended shared ground return is the limit.
Formats	FLAC 16/44.1 and 16/48, MP3	24-bit and higher rates are a stretch goal.
Battery life	≥ 7 h continuous playback	Budget predicts 7.5 h with full derating.
Charge time	~4.5-5 h at 500 mA	Including CV taper. Overnight charging assumed.
Enclosure	~56 x 75 x 13.2 mm	Driven by the 63 x 51 x 6.6 mm cell envelope.

6. Simulation Plan

Scope note. Simulation is applied where it changes design decisions, not as a blanket rule. The power tree's switchers and LDOs qualify: PSRR, output impedance, noise and ripple spectra decide whether the noise-floor target is met. The charger does not: its CC/CV state machine, termination detection, JEITA foldback and PowerPath arbitration are digital behaviours SPICE does not model, and its design work is datasheet arithmetic and thermal layout instead.

6.1 LTspice — boost, charge pump, LDOs

- LDO PSRR versus frequency, focused on 100 kHz - 3 MHz where both switchers' energy lands.
- Output noise spectral density and integrated RMS noise over the audio band.
- LDO output impedance versus frequency — determines rejection of load transients from the DAC and output stage.
- Boost and charge-pump transient behaviour, ripple spectrum, inductor current at worst-case load.
- Startup, UVLO and load-step response, including the SD card's read-burst step.
- Backlight boost ripple and its coupling path, once the display is integrated.

6.2 PSpice for TI / TINA-TI — signal chain

- I/V stage settling, noise contribution, stability against DAC output capacitance.
- MFB filter magnitude, phase, stopband into the MHz region, tolerance sensitivity. Compare against an equivalent Sallen-Key to confirm the stopband argument empirically.
- OPA1622 loop gain and phase margin into capacitive cable loads, with and without output-inclusive compensation.
- Closed-loop output impedance versus frequency — verification of the sub-1-ohm target.
- Gain structure and clipping across the volume range.
- Full-path noise analysis with each contributor identified.

6.3 Hand calculation, not simulation

- Charger inductor ripple and peak current, from the datasheet equations.

- TS network resistor values, from the two simultaneous ratiometric threshold equations.
- Charger junction temperature from theta-JA and dissipation.
- Charge time from capacity, current and CV taper behaviour.

6.4 What simulation will not tell you

SPICE will not model DAC digital behaviour, clock jitter, ground-return coupling, or PCB parasitics. Layout-induced problems are the most likely cause of a disappointing measured result and simulation gives no warning of them.

7. Revision Roadmap

Stage	Hardware	Risk retired	Exit criteria
Rev 0	ESP32-S3 dev kit, off-the-shelf I2S DAC module, 2.4 inch display breakout. No custom PCB.	Firmware	SD mounts over USB MSC; FLAC and MP3 play for an hour with no underrun; minimum buffer occupancy logged; UI navigable.
Rev A	Custom analog board: power tree, ES9038Q2M, I/V, MFB filter, OPA1622 stage. Fed I2S by the dev kit.	Analog + power	Measured output impedance, noise floor and THD+N recorded against simulation. Discrepancies explained, not excused.
Rev B	Integrated 4-layer board: ESP32-S3, storage, USB, BQ24250, full power and analog chain, local oscillators, bare display panel.	Integration	Meets or approaches Section 5 targets. DPLL bandwidth characterised. Runtime and charge behaviour measured. TS thresholds verified against a temperature chamber or equivalent.
Rev C	Mechanical: SolidWorks enclosure, final UI, battery integration.	Usability	Carried and used daily.

Rev A remains the load-bearing decision: splitting the analog board from the MCU board means an analog failure cannot be blamed on firmware or vice versa.

8. Risk Register

Risk	Impact	Mitigation
Battery 28-week lead time	Six-month stall if stock lapses	Purchase two units immediately. The second is needed anyway for discharge characterisation.
No internal NTC in the cell	No hardware charge temperature qualification	Resolved by the BQ24250 TS pin plus an external NTC on the cell body. Moved from firmware responsibility to hardware.
Charge time approaches the safety timer	Charging aborts before completion	Set the 9 h timer via I2C. Do not rely on the 6 h default with a ~4.5-5 h charge.
Charger switching noise near the analog chain	Noise floor degradation while plugged in	3 MHz fundamental is well inside LT3042's PSRR range. Placement, shielded inductor, minimised loop area for the unfilterable magnetic and return-path coupling. Charger is inactive on battery, which is when listening happens.
Charge current at the bottom of range	Reduced margin on the ICHG setting	500 mA is the specified minimum and the datasheet's own design example. LOW_CHG bit forces 330 mA if less is ever wanted.
OPA1622 is 10-pin VSON, thermal pad at V-	Cannot be hand-soldered; pad is at -3.3 V	Hot air or hotplate with a stencil. Dedicated V- pour and via field, never merged into ground.
DC-coupled output fault	Destroys an IEM	Offset detection with a muting device. Required.
ESP32-S3 I2S jitter unquantified	Unknown ceiling on performance	Mitigated architecturally by the DPLL plus a local oscillator; characterise by measurement.
SD read latency causing underrun	Audible dropouts	Size the ring buffer against worst-case latency; log minimum occupancy.
Voltage-based gauging inaccuracy	Misleading battery indicator	Load compensation, measured lookup table, ADC factory calibration. Li-ion discharge curves are deliberately flat mid-range.
USB MSC and playback contending for the volume	Filesystem corruption	State machine: enumeration stops playback and releases the volume before host mount.
Scope creep	Project never finishes	Section 3 is the contract.

9. Open Questions

- **NTC selection** — R25 and beta determine R2 and R3 from the two ratiometric threshold equations. This is the next calculation to close.
- Boost / charge-pump selection: confirm whether LTC3265 or LTC3260 operate directly from SYS or require a pre-boost stage.
- OPA1622 quiescent current at 6.6 V total, read from the Quiescent Current vs. Supply Voltage curve, plus its shutdown current for the standby figure.
- I/V and summing op-amp selection. Quiescent current is a selection criterion — six channels at stake, and the largest controllable load in the budget.
- ES9038Q2M I/V feedback resistor and gain structure, worked backward from the 1 V RMS output target rather than forward from the DAC's reference design.
- ES9038Q2M rail requirements (separate AVCC / DVDD?) and DPLL bandwidth register defaults.
- Volume control method: digital attenuation in the DAC, analog potentiometer, or digital potentiometer ahead of the output stage. Affects noise floor at low listening levels.
- Output muting and DC-fault protection topology.
- Display panel vendor and confirmation that FPC pinout and backlight configuration are published.

- Whether Rev A carries its own power tree or runs from a bench supply.
- Measurement equipment: a -105 dB target needs an analyser floor below it.

Rev 3. Sections 8 and 9 are open by design.